EAS 3530 – Space Systems Concepts

Project 2

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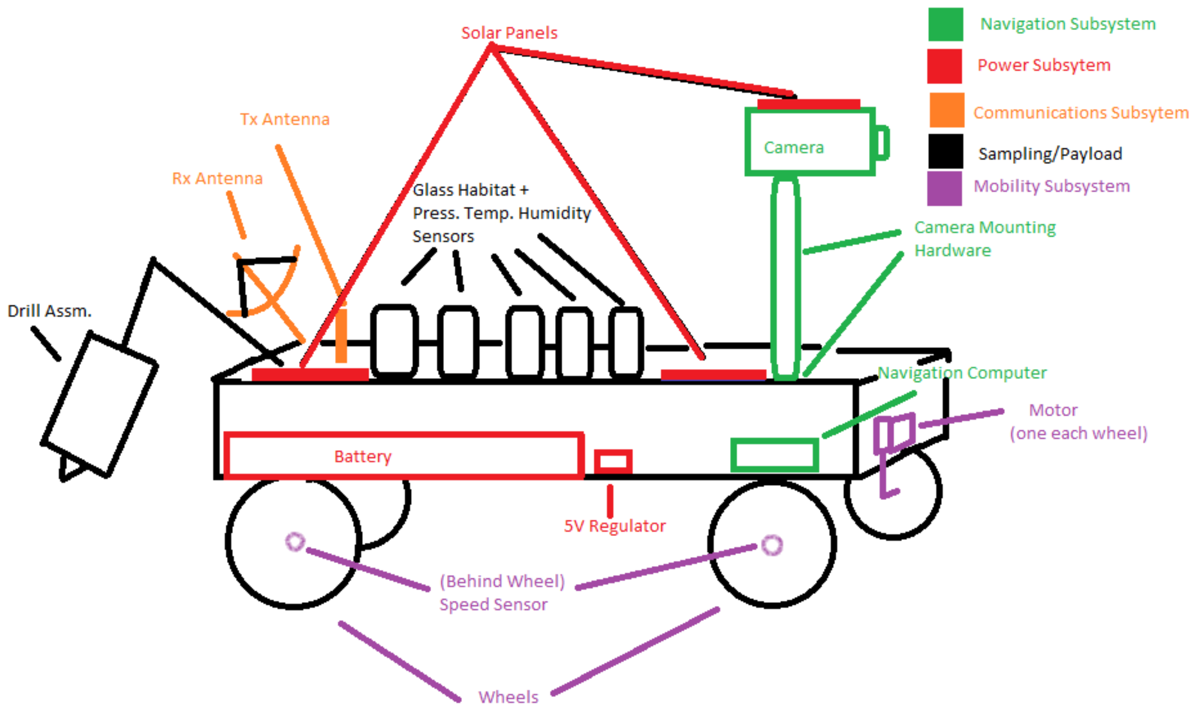
Section 1: Mechanical Subsystem

Sensor	Size	Weight	Power
Hall Effect Speed Sensor	60 mm x 10 mm	20 g	Up to 576 mW
Visual Spectrum Camera	100 mm x 100mm x 70 mm	900 g	20 W
Pressure Sensor (For plant dome)	105 mm x 26 mm	220 g	Up to 450 mW
Temp and Humidity Sensor	95 mm x 15 mm	170 g	1.5 W
Soil Humidity Sensor	94 mm x 24 mm x 75 mm	150g	Up to 240 mW

Subsystem	Component	Size (cm^3)	Mass (kg)	Power Required (W)	QTY
Power					
	Battery (Li+)	70515.303	191.989182		1
	Solar Panel (300W each)	251.34	1		2
	5V Voltage Regulator	2.03	0.05		1
Thermal			0.01	35	1
Communications					
	C&DH, OBC	148.005	0.115	0.2805	1
	Tx Antenna	1622.239906	0.5	80	1
	Rx Antenna	1622.239906	0.5	80	1
Navigation					
	Camera (x2)	700	0.9	20	2
	Navigation Computer	500	0.5	1	1
	Camera Mounting Hardware	5000	20	0	1
Sampling/Payload					
	Drill Assembly	16537.5	8	10	1
	Glass Habitat (x5)	392.6990817	2	0	5
	Pressure, temp, humidity sensors	242.365	0.54	2.5	5

Mobility					
	Speed Sensor	15.24	2.00E-02	5.76E-01	4
	Wheel (x4)	10602.87521	0.735	0	4
	Motor (x4)	356.8849254	0.25	90	4

Subsystem	Component	Total Volume (cm^3)	Total Mass (kg)	Total Power (W)
Power				
	Battery (Li+)	70515.303	191.989182	--
	Solar Panel (300W each)	502.68	2	--
	5V Voltage Regulator	2.03	0.05	--
Thermal			0.01	35
Communications				
	C&DH, OBC	148.005	0.115	0.2805
	Tx Antenna	1622.239906	0.5	80
	Rx Antenna	1622.239906	0.5	80
Navigation			--	--
	Camera (x2)	1400	1.8	40
	Navigation Computer	500	0.5	1
	Camera Mounting Hardware	5000	20	--
Sampling/Payload				
	Drill Assembly	16537.5	8	10
	Glass Habitat (x5)	1963.495408	10	--
	Pressure, temp, humidity sensors	1211.825	2.7	12.5
Mobility				
	Speed Sensor	60.96	8.00E-02	2.30E+00
	Wheel (x4)	42411.50082	2.94	--
	Motor (x4)	1427.539702	1	360
	Totals	144925.3187	242.184182	621.0845



The biggest thing was to ensure that components that had to frequently communicate with one another were close to each other. For example, the camera and the navigation computer will be frequently working with each other, so it is best for them to be near each other. It was also important to keep in mind that the battery would likely take up most of the interior space due to the power requirements of the rover.

Another important decision was whether to place each wheel's motor inside or outside of the main body of the rover. The decision was ultimately made to place it inside, as the lunar dust that could be kicked up while the rover is in motion has the potential to get stuck in seals or moving parts, causing premature failure. A reduction gearbox (which could be used to turn a high RPM motor into a high torque output) can be used to transfer the rotating motion 90 degrees as shown in the diagram.

Section 2: Thermal Control Subsystem

With the total power required while everything is running calculated to be 586 W, we can take that to be Q_{inst}. Next, with the rover being situated on the moon, we can assume the solar constant to be roughly the same as a satellite in Earth orbit (since we are, in a way, orbiting the Earth by being on the moon). This means that $Q_{solar} = 1370\,\frac{\mathrm{W}}{\mathrm{m}^2}$. By the same logic we can also assume the same for both the Earth IR and albedo. We can let the albedo fraction be 0.25 and $Q_{EarthIR} = 239\,\frac{\mathrm{W}}{\mathrm{m}^2}$.

We can write that the total heat input as:

$$Q_{env} = A(Q_{solar}\alpha_{rad} + (\mathrm{Albedo})Q_{solar}\alpha_{rad} + Q_{EarthIR}\epsilon_{rad})$$

Assuming a worst case scenario, we can let the projected area of the rover exposed to the sum be equal to its actual area. To find this area, consider that the total volume calculated in the MEL is about 133000 cm^3, or 0.133 m^3. If we severely idealize, we can imagine this to be a cube of this volume, which would give side lengths of 0.51 meters. This means that the top exposed half of the rover would be 0.51 m x 0.51 m, giving an exposed area of 2606.6 m^2. Assuming an emissivity of 0.86 and the absorptivity of 0.1, we can use the above equation to find that the total environmental heat input during the hot case is 89 watts, and the cold case receives 53.4 watts.

The cold case and hot case would be determined by the lunar day/night cycle, with each lunar day and night lasting about 14 days. During the hot case caused by the lunar day, the craft will be exposed subject to solar irradiance as well as the Earth IR and albedo. During the lunar night's cold case, the only influence will be by Earth IR. Heat would not need to be generated in the hot case, and instead heat pipes or louvres could control how much environmental energy reaches the components to keep them within operational temperatures.

In the cold case, heat will have to be generated. Lunar missions frequently use RHU's to accomplish this (such as on China's Yutu-2 lander), but it should also be possible through passive means (i.e. MLI) to conserve some of that heat from the hot case and focus it on the battery and plant culture. The cold case receives 53.4 watts from the environment as opposed to 89 watts during the hot case. At worst case, we can use heaters to generate the additional 34 watts of useful heating. For this to be added, the battery capacity would have to increase from 64 kWh to 77 kWh to be able to support plant growth throughout the 14-day lunar night.

Component	Operational Temp (^oC)	Survival Temp (^oC)
Battery (Li+)	0 ~ 45	-20 ~ 60
Solar Panel (300W each)	15 ~ 35	-40 ~ 65
5V Voltage Regulator	-10 ~ 40	-40 ~ 125
C&DH, OBC	0 ~ 43	-40 ~ 100
Tx Antenna	--	--
Rx Antenna	--	--
Camera (x2)	-10 ~ 40	-30 ~ 70
Navigation Computer	0 ~ 43	-40 ~ 100
Camera Mounting Hardware	--	--
Drill Assembly	-70 ~ 70	-100 ~ 100
Glass Habitat (x5)	-70 ~ 160	-190 ~ 300
Pressure, temp, humidity sensors	-33 ~ 70	-60 ~ 100
Speed Sensor	-40 ~ 110	-60 ~ 120
Wheel (x4)	--	--

Motors	-40 ~ 120	-70 ~ 150

From this, if we were to ensure that the entire rover was a uniform temperature, the operational temperature would be between 0 and 35 degrees Celsius, and the survival temperature would be between -20 and 60 degrees Celsius.

Flexible heaters could be used for a targeted low-power solution to heating during the cold case. These can provide between 2.5 and 10 watts of heat per square inch and are very low profile and take up almost no volume, on top of being very efficient.

Section 3: Electrical Power Subsystem

Power Modes	Survival	Moving	Sampling	Plant Growing
C&DH, OBC	Y	Y	Y	Y
Tx Antenna	N	Y	Y	Y
Rx Antenna	Y	N	Y	Y
Camera	N	Y	N	N
Navigation Computer	N	Y	N	N
Drill Assembly	N	N	Y	N
PT&H sensors	N	Y	Y	Y
Speed Sensor	N	Y	N	N
Motors	N	Y	N	N
Thermal	Y	Y	Y	Y

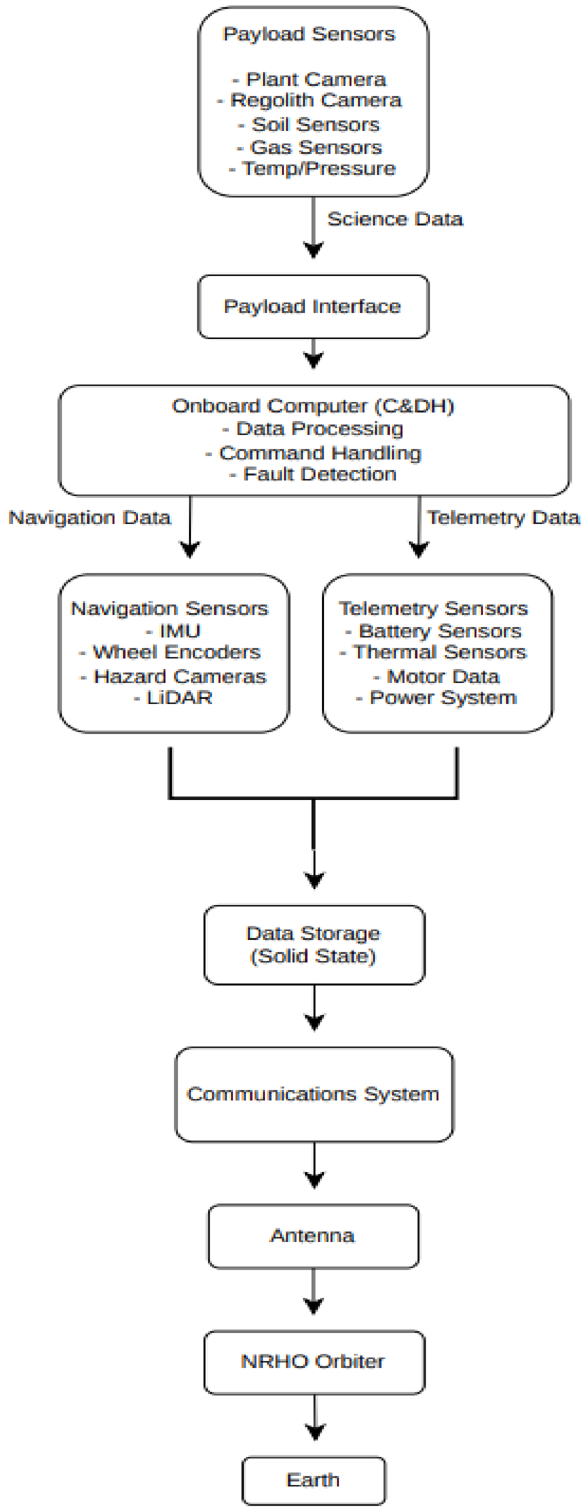
Deployable solar arrays extending out from the sides of the rover should be enough to generate sufficient power. A tradeoff had to be made between being power positive in the most demanding power mode and the total size and weight of the solar array. It was decided that, since most of the time will be spent on growing the plant, the rover should be power positive in this scenario. While this does mean that the rover will be power negative while moving, since moving will only be happening for short periods of time relative to the rest of the mission, it should not be of major concern.

To charge the 74-kWh battery from zero to full capacity within the 14 days the rover has in the sun, we need about 450 watts of useful power generation. With a high-efficiency panel of about 300 W/m^2, we can achieve this with about 1.5 m^2 of solar panel area after accounting for losses for power conversion inherent in the design of a solar panel. Three panels placed on all sides except the front would be best, each with an area of about 0.5 m^2, providing about 150 watts each. If the lander is on the lunar south pole, it is best for these panels to be facing almost perpendicular to the ground, so that the panels will always be facing the sun.

Power Modes	Peak Power	Survival	Moving	Sampling	Plant Growing
C&DH, OBC	**0.2805**	**0.2805**	**0.2805**	**0.2805**	**0.2805**
Tx Antenna	**80**	**Off**	**80**	**80**	**80**
Rx Antenna	**80**	**80**	**Off**	**80**	**80**
Camera	**40**	**Off**	**40**	**Off**	**Off**
Navigation Computer	**1**	**Off**	**1**	**Off**	**Off**
Drill Assembly	**10**	**Off**	**Off**	**10**	**Off**
PT&H sensors	**12.5**	**Off**	**12.5**	**12.5**	**12.5**
Speed Sensor	**2.3**	**Off**	**2.3**	**Off**	**Off**
Motors	**360**	**Off**	**360**	**Off**	**Off**
Thermal	**35**	**35**	**35**	**35**	**35**
Power Needed (W)	**621.0805**	**115.2805**	**531.0805**	**217.7805**	**207.7805**

While the rover would not be power positive in all modes, it would be positive in the modes where most of the mission time is spent. It only fails to be power positive in the peak power situation, which would never realistically happen as it requires a sample to be taken while in motion, or while the rover is moving from one location to another. To compensate for this, the time while the rover is in motion should be minimized such that the batteries can be charged for most of the time during the night cycle. With the solar panels creating 450W, it should be able to charge the battery from zero to 100% in about 13 days if it is in the sampling or plant growing phase.

Section 4: C&DH Subsystem



- **Estimated Payload Sensor Data:**

Sensor	Assumption	Estimated Data Rate
Plant growth Camera	1 compressed image every 10 minutes	0.01 Mbps
Regolith/sample camera	Short image bursts during sampling	0.05 Mbps average
Spectrometer/Chemical Sensor	Periodic Measurements	0.05 Mbps
Soil moisture sensor	Low-rate numerical data	0.001 Mbps
Temperature sensor	Low-rate numerical data	0.001 Mbps
Pressure sensor	Low-rate numerical data	0.001 Mbps
Gas sensor	Low-rate numerical data	0.001 Mbps
Growth Chamber status	Low-rate event data	0.001 Mbps
Total Mbps:		**0.12 Mbps**

- **Estimated Navigation and Rover Data:**

Sensor	Assumption	Estimated Data Rate
Navigation Cameras	Compressed images for terrain awareness	0.5 Mbps
LiDAR/Terrain mapping	Periodic terrain scans	0.5 Mbps
IMU	Continuous low-rate data	0.01 Mbps
Wheel encoders	Continuous low-rate data	0.005 Mbps
Motor telemetry	Current, speed, temperature	0.005 Mbps
Rover Health telemetry	Power, thermal, comms status	0.01 Mbps
TOTAL Mbps:		**1.03 Mbps**

- **Final Estimate:**
  - **1.15 Mbps** raw data rate

  This is close to Project 1 communications requirements, which state that the communication system should transmit telemetry at about 1 Mbps

- Processor Throughput Estimate:

The onboard computer must support:

- Sensor Collection
- Image Compression
- Navigation processing
- Terrain hazard detection
- Mobility control
- Data packet formatting
- Fault detection

- Communications management

A reasonable estimate is:

Task	Estimated Processing Load
Payload sensor handling	Low
Image compression	Moderate
Navigation and obstacle detection	Moderate to high
Rover mobility control	Moderate
Communications packet handling	Low to moderate
Fault detection and recovery	Moderate

- **For this mission, the processor should support at least:**
  - **200 to 400 MIPS**
  - A radiation hardened flight computer similar to a **RAD750-class processor** would be appropriate. The design could also use a lower power microcontroller for basic rover control and a separate processor for image/navigation tasks.

Data Collected Per Week:

$$1.15 \text{ Mbps} = \frac{1.15}{8} \quad \text{MB/s}$$

= **0.14375 MB/s**

Data per day:

86,400 seconds/day

0.14375 MB/s * 86,400 seconds/day = 12,420 MB/day

= **12.42 GB/day**

Data per week:

12.42 Gb/day * 7 days = **86.94 GB/week**

- Project 1 listed a 60 day rover operating requirement for the mobility system. If the rover stores all raw data for 60 days:

12.42 GB/day * 60 days

= **745.2 GB (Total raw mission data)**

With a 25% margin:

$$745.2 \text{ GB} * 1.25$$

$$= \mathbf{931.5 \text{ GB (Recommended storage)}}$$

Storage Strategy:

- The rover should not rely on transmitting every raw data file immediately. Instead, it should:
  - Store all critical data locally.
  - Compress image and terrain data
  - Prioritize transmission of science data.
  - Delete redundant navigation imagery after successful downlink
  - Keep summary telemetry and key science results for the full mission
- The recommended onboard storage would be a 1 TB radiation tolerant solid-state storage
  - This gives enough margin to store full mission data if communication windows are reduced or if the rover needs to operate without immediate downlink.

Section 5: Telecommunications Subsystem

Rover:

- The Rover should use a medium gain patch antenna or low profile phased patch antenna because:
    - Compact
    - Low mass
    - No large deployment mechanism
    - Better suited to a rover than a large dish
    - Can maintain communication with the orbiter over a wide range of pointing angles

Orbiter:

- The NRHO orbiter should use a high gain parabolic dish antenna for Earth communication.
    - Earth Moon distance is large
    - High data rates require high antenna gain
    - A dish antenna improves downlink margin
    - Orbiter has better pointing stability than the rover

Ground Station Antenna:

- Project 1 listed ground stations at Cape Canaveral, the United Kingdom, and Guam to maintain high daily communication coverage

- X-Band (8-12 GHz)
    - This is the NASA Standard
    - Good balance: bandwidth + lower atmospheric loss
    - Works with deep space networks
    - Higher data capability than UHF/S-band
    - BER = 10^{-5}

- $\frac{E_b}{N_0} = 9.6\ dB$

Link Budget:

Prompt Segment	Downlink: Orbiter --> Ground Station	Uplink: Ground Station --> Orbiter
Transmitter power from power budget	20 W = 13 dBW	100 W = 20 dBW
Tx line loss	0.3 dB	0.3 dB
Rx line loss	0.3 dB	0.3dB
Frequency band	X-band, 8.4 GHz	X-band, 7.2 GHz
Distance / range	400,000 km	400,000 km
Space loss	222.98 dB	221.64 dB
Atmospheric loss	1 dB	1 dB
Data rate	500 kbps	10 kbps
System temperature	500K	500K
Spacecraft antenna gain	35 dBi	35 dBi
Ground station antenna gain	60 dBi	60 dBi
Required Eb/No for BER 10^{-5}	9.6 dB	9.6 dB
Calculated Eb/No	28.04 dB	52.37 dB
Link margin	18.44 dB	42.77 dB

Iteration	Tx Gain (dBi)	Rx Gain (dBi)	Data Rate	Eb/No (dB)	Margin (dB)	Result
1 (initial)	25	50	1 Mbps	~8 dB	-1.6 dB	Link does not close

2	35	50	1 Mbps	~18 dB	+8.4 dB	Works but low margin
3 (Final)	35	60	500 kbpa	28.04 dB	+18.4 dB	Strong margin

- The initial design used a 25 dBi spacecraft antenna and a 50 dBi ground station antenna with a Mbps data rate. This resulted in insufficient link margin, and the link did not close
- In the second iteration, the spacecraft antenna gain increased to 35 dBi, which improved performance but still resulted in a relatively low margin.
- In the final iteration, the ground station antenna gain was increased to 60 dBi and the average data rate was reduced to 500 kbps. This resulted in a final link margin of approximately 18 dB, ensuring reliable communication

- Yes, the required ground antenna gain of 60 dBi is consistent with large deep-space ground stations. Since project 1 selected ground stations at Cape Canaveral, the United Kingdom, and Guam, the mission can maintain high daily communication coverage through geographically separated ground stations.

Section 6: General Thoughts

Meeting all subsystem requirements was challenging due to the strong interdependence between systems. The payload required multiple sensors and cameras, which increased data generation rate. This directly impacted the C&DH subsystem, which required higher processing capability and storage. In turn, this increased the load on the telecommunications subsystem, which needed to support higher data rates, and the power subsystem, which needed to supply sufficient energy for transmission.

One requirement that could be reconsidered is the rover speed requirement of at least 1 m/s. For a science focused mission involving regolith collection and plant growth experiments, high speed traversal is not critical. Reducing this requirement would decrease power consumption, simplify mobility design, and reduce stress on the system.

Additionally, the requirement to transmit large amounts of terrain mapping data could be relaxed. Prioritizing only essential science data would reduce communications demands and simplify the overall systems of design.

The most difficult budget to close was the telecommunications link budget.

This is because:

- The Earth-Moon distance creates extremely high signal loss
- The rover generates a significant amount of data
- The spacecraft has limited transmit power
- Antenna size is constrained by mass and volume

To overcome this, the system required:

- A relay orbiter in NRHO
- A high gain antenna
- Reduced data rate
- Data compression strategies

- Multiple iterations were required.
    - Data rates were reduced after initial estimates were too high
    - Antenna size was increased to improve link margin
    - Power system constraints required limiting transmission time
    - Storage strategy was updated to avoid transmitting all raw data